Instructions:

1. This question paper contains a total of **1** page (**1** side of paper).
2. Write your name, roll number, department, and section on every side of every sheet of this booklet
3. Write final answers neatly with a blue/black pen in the given boxes.
4. **Answers written outside the box will NOT be graded.**

Total **10 Marks**

Q. 1: Write the output of the following program in the appropriate box and answer the question. (2+2+2 = 6 Marks)

```
#include<stdio.h>
void func1(int *arr){
  for(int i=1; i<=5;i++)
    (i-1)[arr] = (i%5);
}
void func2 (int a[1] , int arr){
  a[0] = arr + 3;
}
int main(){
  int arr[6] = {2,2,2,2,2};
  func1(arr+1);
  for(int i=0; i<6; i++)
        printf("%d ", i[arr] );
  printf("\n");
  func2(arr, *arr);
  for(int i=0; i<6; i++)
        printf("%d ", i[arr] );
  return 0;
}
```

Output

2 1 2 3 4 0

5 1 2 3 4 0

Will the answer change if we write func2 as:

```
void func2 (int a[1] , int arr){
 int temp = arr + 4;
 a = &temp;
}
```

Explain your answer

yes, in this case the 2nd output will be different. It will be 2 1 2 3 4 0. It is because in func2, only the memory location that points of a changes, not the value it was pointing to.

Q. 2: Write the output of the following sequence of instructions **long arr2[90]; int arr1[80], a1=(&arr2[60]), a2=(&arr2[10]), b1=(&arr1[60]), b2=(&arr1[10]); printf("%d %d", a1-a2 , b1-b2);** Explain your answer. (2+2 Marks)

O/P: 400 200

Explanation: a_1 = arr2 + 60*size of long. a_2 = arr2 + 10*size of long. $a_2 - a_1$ = 50* size of long = 50*8 = 400. Similarly, b_1 = arr1 + 60* size of int. b_2 = arr1 + 10* size of int. $b_1 - b_2$ = 50* size of int = 50*4 = 200.